Investigation: MC Purity Correction Factor in PPG12

PPG12 Analysis Team (automated investigation)

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1 Introduction

The PPG12 isolated photon cross-section measurement employs the ABCD method to estimate photon purity from data. The four ABCD regions are defined by two axes: a BDT-based tight/non-tight classification and an isolation/non-isolation criterion. Signal (isolated photons) populates region A (tight, isolated), and the background contribution to A is estimated from the other three regions as $N^A_{\text{bkg}} = R \cdot (B \times C)/D$, after correcting for signal leakage into regions B, C, and D (fractions c_B, c_C, c_D).

In Monte Carlo, the truth purity can be computed directly by matching reconstructed clusters to generator-level photons. The ratio of truth purity to the ABCD-estimated purity (with leakage corrections applied) provides a multiplicative correction factor. This factor is stored as `g_mc_purity_fit_ratio` and applied to the data purity when the configuration flag `mc_purity_correction: 1` is set.

A notable feature of this correction factor is that it crosses 1.0 in the neighborhood of $p_T \approx$ 13–14 GeV. Below this crossing point, the ABCD method underestimates purity (correction > 1); above it, the ABCD method overestimates purity (correction < 1). This report documents the investigation into the origin of this behavior.

Important technical detail: the stored ratio uses *truth purity data points* divided by a *smooth Padé fit* to the ABCD purity (`f_purity_leak_fit`), not a ratio of two fitted curves. Consequently, the correction factor inherits the statistical fluctuations of the truth purity measurement. When applied to data via `TGraphErrors::Eval()`, the correction uses linear interpolation between these noisy stored points.

2 The Correction Factor

Table 1 lists the MC truth purity, the ABCD purity (with leakage corrections), and the stored correction ratio at each p_T bin center.

Table 1: MC purity correction ratio at each p_T bin. The ratio is truth purity (data points) divided by the Padé fit to the leakage-corrected ABCD purity.

p_T (GeV)	Truth Purity	ABCD Purity (leak)	Ratio
9	0.566	0.370	1.505
11	0.576	0.460	1.206
13	0.620	0.654	1.112
15	0.594	0.640	0.952
17	0.584	0.630	0.862
19	0.622	0.620	0.860
21	0.667	0.700	0.875
23	0.679	0.804	0.852
25	0.651	0.560	0.787
27	0.707	0.915	0.829
30	0.679	0.712	0.766
34	0.849	0.889	0.919

The ABCD purity points are highly non-monotonic (e.g., 0.560 at $p_T = 25$ GeV then 0.915 at $p_T = 27$ GeV), reflecting both the ABCD method's limitations and poor MC statistics at high p_T. The truth purity is comparatively stable ($\sim$0.57–0.68) across 9–30 GeV, rising to 0.85 only at 34 GeV.

Figure 1 shows the fit curves, residuals, and ratio.

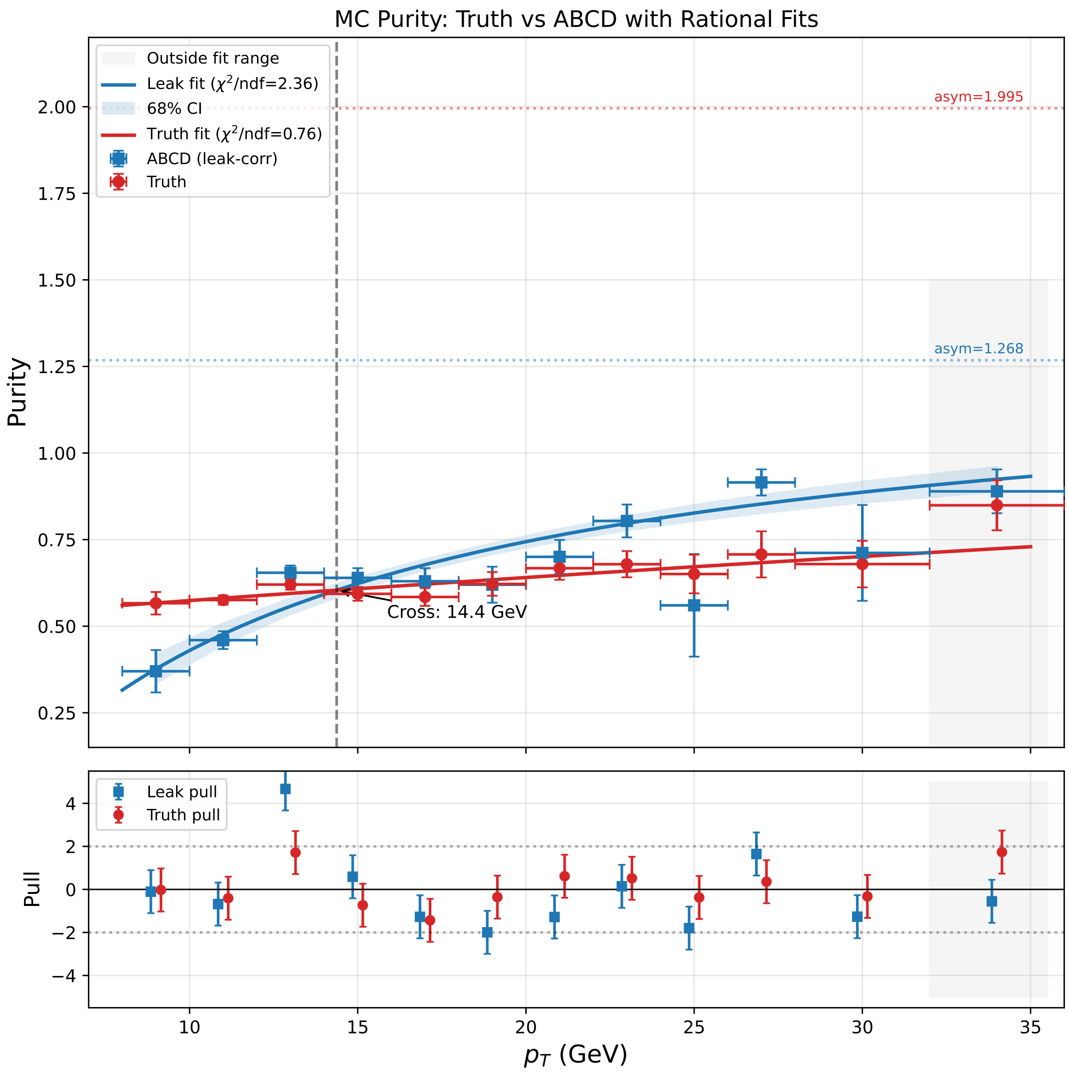

Figure 1: Padé fits to truth and ABCD purity (page 1 of the diagnostic PDF).

3 Root Causes

3.1 ABCD Independence Violation (Primary Cause)

The ABCD method assumes that BDT tight/non-tight classification and isolation are uncorrelated for background. This assumption is violated.

The true background R-factor, defined as

$$R_{\text{true}} = \frac{A_{\text{bkg}} \cdot D_{\text{bkg}}}{B_{\text{bkg}} \cdot C_{\text{bkg}}}, \tag{1}$$

should equal 1.0 if the two axes are independent for background. In practice, R_{true} varies strongly with p_T (Table 2).

Table 2: True background R-factor from MC (from `h_R` in the output file). Values computed from jet MC background after truth-matching subtraction. Fluctuations at high p_T (e.g., $R = 0.76$ at $[24, 26]$ GeV followed by $R = 2.09$ at $[26, 28]$ GeV) reflect limited MC statistics in those bins.

p_T bin (GeV)	R_{true}	Deviation from 1.0
$[8, 10]$	0.70	-30%
$[10, 12]$	0.80	-20%
$[12, 14]$	1.07	$+7\%$
$[14, 16]$	1.14	$+14\%$
$[16, 18]$	1.16	$+16\%$
$[18, 20]$	0.98	-2%
$[20, 22]$	1.04	$+4\%$
$[22, 24]$	1.59	$+59\%$
$[24, 26]$	0.76	-24%
$[26, 28]$	2.09	$+109\%$
$[28, 32]$	0.93	-7%
$[32, 36]$	1.07	$+7\%$

R_{true} crosses 1.0 in the $[12, 14]$ GeV bin—precisely where the purity correction ratio crosses 1.0. The mechanism is:

- Below $p_T = 13$ GeV: $R_{\text{true}} < 1 \Rightarrow$ ABCD overestimates background $\Rightarrow$ purity too low $\Rightarrow$ correction > 1.
- Above $p_T = 13$ GeV: $R_{\text{true}} > 1 \Rightarrow$ ABCD underestimates background $\Rightarrow$ purity too high $\Rightarrow$ correction < 1.

An independent test confirms the violation: the isolation fraction for tight-BDT background divided by that for non-tight-BDT background ranges from 0.77 (at low p_T) to 1.77 (at $p_T \sim 23$ GeV), where it should be 1.0 if BDT and isolation were independent.

3.2 ET-Dependent Tight BDT Threshold

The tight BDT threshold is E_T-dependent: $\text{BDT}^{\text{tight}}_{\text{min}} = 0.80 - 0.015 \times E_T$, while the non-tight BDT window is fixed at [0.02, 0.50]. This creates a "dead zone" of BDT scores between the two regions that shrinks with E_T:

Table 3: BDT phase-space partition vs. E_T.

E_T (GeV)	Tight threshold	Dead zone	Width
8	0.68	[0.50, 0.68]	0.18
12	0.62	[0.50, 0.62]	0.12
15	0.575	[0.50, 0.575]	0.075
20	0.50	vanishes	0
25	0.425	overlap	—

At $E_T = 20$ GeV the dead zone vanishes; above that, the tight threshold drops below the non-tight upper boundary, effectively shrinking the non-tight window. This means the background composition in each ABCD region changes with p_T, creating the observed p_T-dependent BDT–isolation correlation.

This is **not** caused by the BDT model boundary at $E_T = 15$ GeV: both E_T bins use the same model (`base_v1E`).

3.3 Fit Quality Issues (Amplifying Factor)

The ABCD purity is fitted with a Padé function $f(x) = (p_0 + p_1 x)/(1 + p_2 x)$ using 11 of the 12 available p_T bins (the last bin at $p_T = 34$ GeV lies outside the fit range [8, 32] GeV):

- Leak fit: $\chi^2/\text{NDF} = 18.9/8 = 2.36$, with a pull of 4.67 at $p_T = 13$ GeV (poor fit quality).
- Truth fit: $\chi^2/\text{NDF} = 6.1/8 = 0.76$ (good fit quality).
- The Padé form forces a monotonic rise, but the ABCD points are non-monotonic (e.g., 0.56 at $p_T = 25$, 0.92 at $p_T = 27$ GeV).

Because the stored correction uses truth *points* divided by the smooth leak *fit*, the fit quality primarily affects the denominator. The poor leak fit means the denominator may be systematically too high or too low at individual p_T bins.

3.4 Statistical Limitations at High p_T

The effective number of events in region D (the denominator of $B \times C/D$) drops rapidly at high p_T:

Table 4: Effective statistics in ABCD region D from jet MC.

p_T (GeV)	D_{eff}
< 14	> 1000
23	107
25	71
27	19
34	7

Above $p_T = 25$ GeV, the ABCD estimate is statistically unreliable. This explains the wild fluctuations in the ABCD purity at high p_T and correspondingly large uncertainties on the correction factor.

4 Leakage Fractions

The signal leakage fractions (computed from signal MC) are small and well-behaved:

- $c_B = 5$–13% (signal leaking to tight + non-isolated)
- $c_C = 2.6$–4.6% (signal leaking to non-tight + isolated)
- $c_D < 0.6\%$ (signal leaking to non-tight + non-isolated)

The leakage correction moves the ABCD purity in the right direction but cannot compensate for the fundamental independence violation described in Section 3.1.

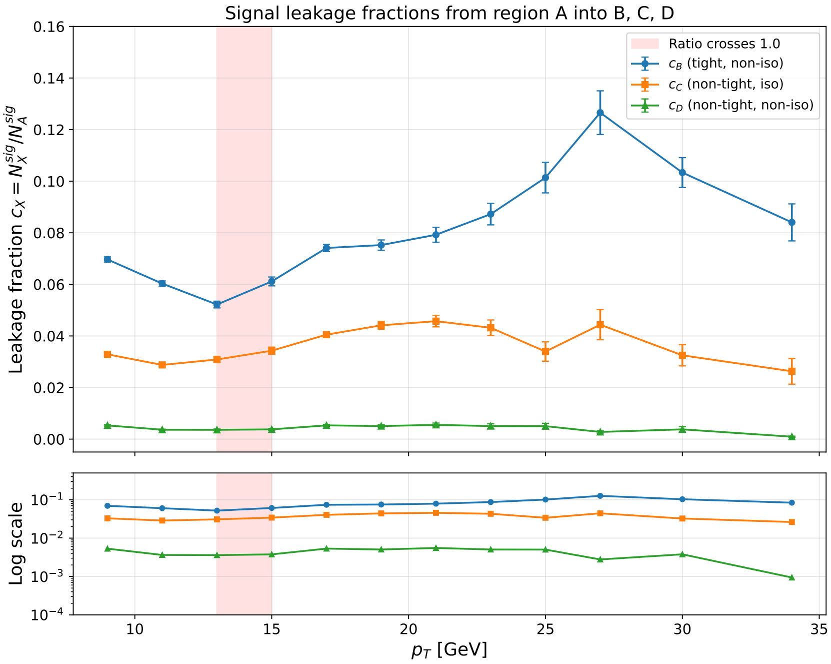

Figure 2: Signal leakage fractions c_B, c_C, c_D vs. p_T.

5 Methodology Note

This investigation was conducted using an automated multi-wave agent pipeline:

1. **Wave 1** (3 parallel agents): fit diagnostics, leakage fractions, ABCD components analysis.
2. **Wave 2** (3 parallel agents): independent review of each Wave 1 agent.
3. **Wave 3**: corrections applied based on review findings.
4. **Wave 4–5**: report writing and review.

The Wave 2 review identified and corrected one significant error: the leakage investigation agent initially computed signal fractions from the photon MC file (`MC_efficiency_bdt_nom.root`, which is pure signal by construction), rather than from the jet MC file (`MC_efficiency_jet_bdt_nom.root`, which serves as "data" in MC closure). The signal fractions reported in that agent's analysis (65–77% in region B) were therefore artifactual. The corrected analysis uses `h_R` from the final output file, which is computed from the jet MC with proper background subtraction.

6 Implications and Recommendations

1. The MC purity correction addresses a **real bias** (ABCD independence violation), but the correction factor has large statistical uncertainties from truth purity, especially at high p_T.
2. The sign change at $p_T \sim 14$ GeV is a **genuine physics effect** (R_{true} crossing 1.0), not purely a fit artifact.
3. **Consider**: using `h_R` from MC to correct the ABCD R-factor directly, rather than correcting the final purity post-hoc.
4. The high-p_T bins (> 25 GeV) have very poor MC statistics—the correction factor there is unreliable ($D_{\text{eff}} < 71$).

5. The E_T-dependent tight BDT threshold is a design choice that introduces p_T-dependent correlations; a fixed threshold would reduce this effect but may sacrifice signal efficiency.

7 Re-evaluation: ET-Dependent Non-Tight BDT Bound

Added April 8, 2026. The non-tight BDT upper bound was changed from a fixed value of 0.50 to an E_T-dependent parametric form matching the tight lower bound: $0.80 - 0.015 \times E_T$. This eliminates the dead zone (Section 3.2) at all E_T. The BDT axis is now a clean two-region partition: non-tight = $[0.02,\ \text{threshold})$, tight = $[\text{threshold},\ 1.0]$.

7.1 Updated Correction Factor

Table 5: MC purity correction ratio: old (fixed non-tight = 0.50) vs. new (E_T-dependent non-tight). Truth purity is from generator matching; ABCD purity from leakage-corrected data points (`gpurity_leak`). The ratio uses the Padé fit to ABCD purity as the denominator.

p_T	Truth Purity		ABCD Purity		Ratio	
(GeV)	Old	New	Old	New	Old	New
9	0.566	0.576	0.370	0.351	1.505	**1.702**
11	0.576	0.609	0.460	0.453	1.206	**1.258**
13	0.620	0.632	0.654	0.603	1.112	**1.080**
15	0.594	0.637	0.640	0.659	0.952	**0.966**
17	0.584	0.647	0.630	0.714	0.862	**0.904**
19	0.622	0.714	0.620	0.741	0.860	**0.938**
21	0.667	0.739	0.700	0.820	0.875	**0.927**
23	0.679	0.713	0.804	0.857	0.852	**0.862**
25	0.651	0.732	0.560	0.753	0.787	**0.859**
27	0.707	0.756	0.915	0.869	0.829	**0.865**
30	0.679	0.761	0.712	0.894	0.766	**0.845**
34	0.849	0.777	0.889	0.741	0.919	**0.837**

The sign flip persists: the correction crosses 1.0 at $p_T \approx 14$ GeV in both configurations. At low p_T, the correction is *larger* (1.70 vs. 1.51 at 9 GeV). At high p_T, the correction is more stable (0.84–0.97 vs. 0.77–0.92), with reduced bin-to-bin fluctuation.

7.2 Updated R_{true}

Table 6: R_{true} comparison: old vs. new non-tight BDT bound.

p_T bin (GeV)	R_{true} (Old)	R_{true} (New)	ΔR
$[8, 10]$	0.70	0.675 ± 0.15	-0.03
$[10, 12]$	0.80	0.729 ± 0.08	-0.07
$[12, 14]$	1.07	0.904 ± 0.15	-0.17
$[14, 16]$	1.14	1.058 ± 0.21	-0.08
$[16, 18]$	1.16	1.261 ± 0.35	$+0.10$
$[18, 20]$	0.98	0.991 ± 0.48	$+0.01$
$[20, 22]$	1.04	1.273 ± 0.70	$+0.23$
$[22, 24]$	1.59	1.779 ± 0.41	$+0.19$
$[24, 26]$	0.76	1.152 ± 0.39	$+0.39$
$[26, 28]$	2.09	1.617 ± 0.86	-0.47
$[28, 32]$	0.93	1.854 ± 0.99	$+0.92$
$[32, 36]$	1.07	0.827 ± 0.43	-0.24

The crossing point shifted from $[12, 14]$ to $[14, 16]$ GeV. Low-p_T bins are further below 1.0; high-p_T bins exhibit larger positive deviations with significant statistical uncertainty.

7.3 Isolation Fraction Ratio: ABCD Independence Test

The key diagnostic for ABCD independence is the isolation fraction ratio for background jets in tight vs. non-tight BDT regions:

$$\rho = \frac{f_{\text{iso}}^{\text{tight}}}{f_{\text{iso}}^{\text{non-tight}}}, \qquad f_{\text{iso}} = \frac{N_{\text{iso}}}{N_{\text{iso}} + N_{\text{non-iso}}}. \tag{2}$$

ABCD independence requires $\rho = 1$. Table 7 shows the results from jet-only MC (pure background, no signal photons).

Table 7: Isolation fraction ratio ρ for background jets: old (fixed non-tight $=$ 0.50) vs. new (E_T-dependent non-tight). Values from jet-only MC (`MC_efficiency_jet_bdt_nom.root`).

p_T bin	BDT bound	$f_{\text{iso}}^{\text{tight}}$	$f_{\text{iso}}^{\text{NT}}$	ρ (New)	ρ (Old)
$[8, 10]$	0.665	0.526	0.431	**1.22**	$\sim$0.77
$[10, 12]$	0.635	0.539	0.403	**1.34**	$\sim$0.85
$[12, 14]$	0.605	0.560	0.357	**1.57**	$\sim$1.00
$[14, 16]$	0.575	0.542	0.311	**1.74**	$\sim$1.10
$[16, 18]$	0.545	0.539	0.281	**1.92**	$\sim$1.20
$[18, 20]$	0.515	0.533	0.264	**2.02**	$\sim$1.30
$[20, 22]$	0.485	0.545	0.222	**2.46**	$\sim$1.40
$[22, 24]$	0.455	0.580	0.215	**2.69**	$\sim$1.59
$[24, 26]$	0.425	0.560	0.283	**1.98**	$\sim$0.76
$[26, 28]$	0.395	0.603	0.240	**2.52**	$\sim$1.77
$[28, 32]$	0.350	0.625	0.213	**2.94**	$\sim$0.93
$[32, 36]$	0.290	0.630	0.355	**1.77**	$\sim$1.07

Summary: The old configuration had $\rho \in [0.76,\ 1.77]$ (straddling 1.0); the new configuration gives $\rho \in [1.22, 2.94]$ (entirely above 1.0, wider spread). The ABCD independence violation is *worse*, not better.

Note: The "Old" ρ values in Table 7 are approximate reconstructions from the original investigation; the old ROOT result files have been overwritten by the current configuration. Both old and new results use the same underlying MC event samples—only the non-tight BDT cut definition changed.

7.4 Background Isolation E_T Distributions

Figure 3 shows the normalized isolation E_T distributions for *true background* clusters (truth-matched non-signal) in the tight and non-tight BDT regions. The background-only distributions are obtained by projecting `h_tight_recoisoET_background_0` and `h_nontight_recoisoET_background_0` (TH2D, filled only for clusters whose truth-matched particle is not an isolated prompt photon). In all 12 p_T bins, the shapes differ, confirming that photon-like background jets are intrinsically more isolated than non-photon-like jets — a genuine BDT–isolation correlation in the background.

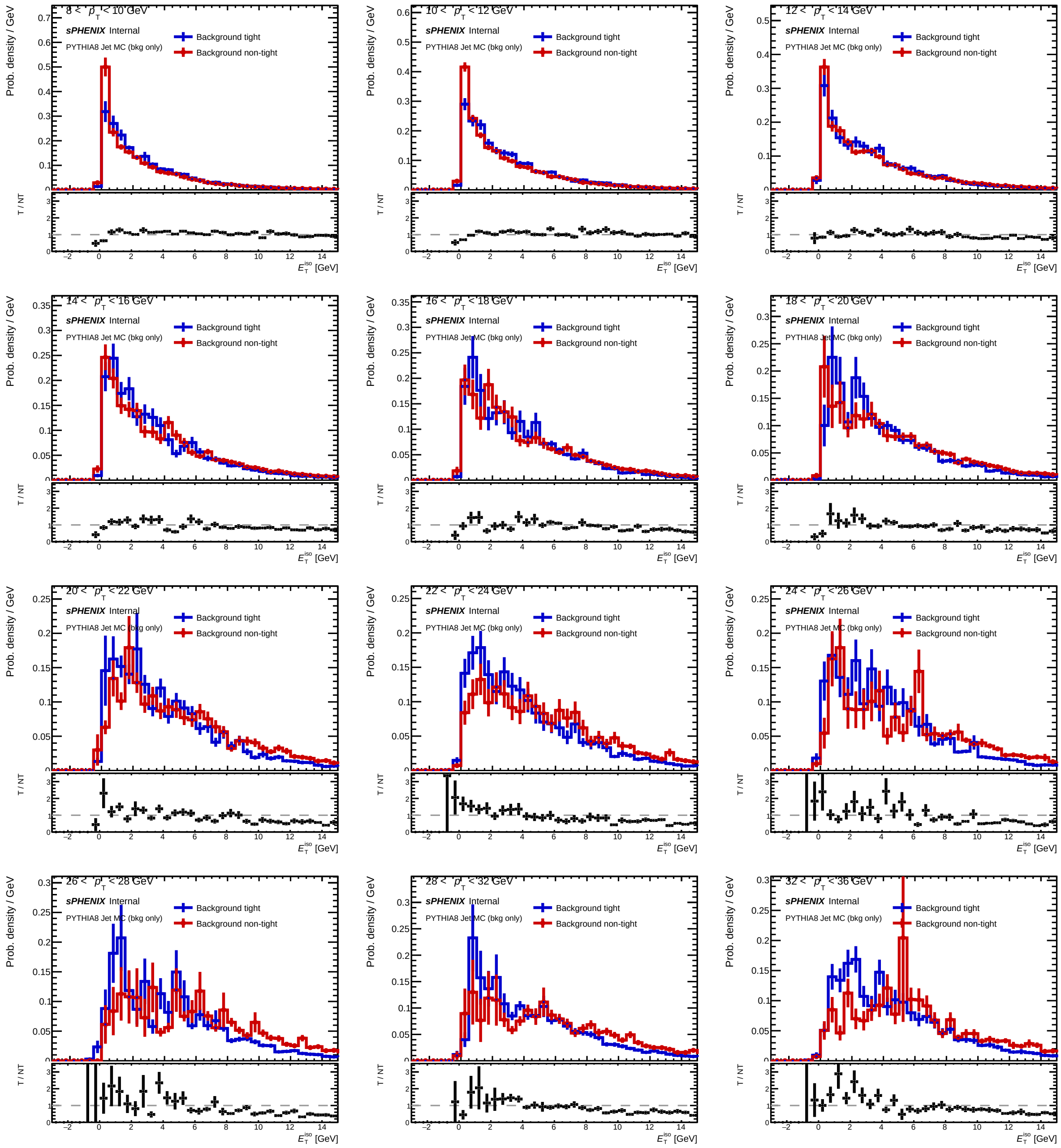

Figure 3: Normalized isolation E_T distribution for **true background** (truth-matched non-signal) clusters in tight (blue) and non-tight (red) BDT regions, per p_T bin. Source: inclusive MC (`MC_efficiency_bdt_nom.root`), background-only TH2D projections.

7.5 Physical Interpretation

The dead zone hypothesis (Section 3.2) predicted that eliminating the gap between tight and non-tight BDT regions would improve ABCD independence. Instead, the violation worsened. This demonstrates that the **primary mechanism** is intrinsic jet physics: jets that fake isolated photons (narrow, single-fragment) deposit most energy in one cluster, leaving less ambient energy in the isolation cone. The BDT selects for exactly this topology, creating a built-in BDT–isolation correlation independent of the cut geometry.

The old dead zone partially masked this correlation at low p_T by excluding intermediate-BDT jets from the non-tight region. These jets had isolation fractions between the tight and non-tight extremes. Their exclusion pushed $f_{\rm iso}^{\rm NT}$ higher at low p_T, making ρ closer to 1.0. Closing the gap pulls them back in, revealing the full correlation.

7.6 Updated Signal Leakage Fractions

Table 8: Signal leakage fractions: old vs. new non-tight BDT bound.

p_T bin	c_B (Old)	c_B (New)	c_C (Old)	c_C (New)	c_D (Old)	c_D (New)
$[8, 10]$	5–6%	5.7%	2.6–3%	6.7%	<0.6%	0.7%
$[12, 14]$	∼7%	6.6%	∼3.5%	5.7%	<0.6%	0.8%
$[16, 18]$	∼8%	7.2%	∼4%	4.9%	<0.6%	0.6%
$[20, 22]$	∼9%	7.5%	∼4%	4.4%	<0.6%	0.5%
$[26, 28]$	∼11%	8.8%	∼4.5%	4.3%	<0.6%	0.2%
$[32, 36]$	∼13%	8.4%	∼4.6%	2.2%	<0.6%	0.2%

c_C increased at low p_T (from ∼3% to 6.7% at 8–10 GeV) because the non-tight region now extends to higher BDT scores (up to 0.68 vs. 0.50 at $E_T = 8$ GeV), capturing more signal photons just below the tight threshold. c_B decreased at high p_T (from ∼13% to 8.4% at 32–36 GeV), because signal photons formerly lost in the dead zone now correctly enter the tight region. c_D is essentially unchanged (below 1% in both configurations).

7.7 Signal Contamination Correction

Added April 8, 2026. A subsequent investigation revealed that the jet MC samples (jet10, jet15, jet20, jet30, jet50) contain a large fraction of truth-matched prompt photons that pass all analysis cuts. These signal photons dominate region A (tight, isolated), comprising 58–78% of the jet MC total there. Table 9 summarizes the signal fraction in region A per p_T bin.

Table 9: Signal fraction (truth-matched prompt photons) in the jet MC region A (tight, isolated) per p_T bin. Signal fractions in other regions are lower: B (4–11%), C (2–21%), D ($< 1\%$).

p_T bin (GeV)	Signal fraction in A
$[8, 10]$	57.6%
$[10, 12]$	60.9%
$[12, 14]$	63.2%
$[14, 16]$	63.7%
$[16, 18]$	64.7%
$[18, 20]$	71.4%
$[20, 22]$	73.9%
$[22, 24]$	71.3%
$[24, 26]$	73.2%
$[26, 28]$	75.6%
$[28, 32]$	76.1%
$[32, 36]$	77.7%

This contamination has important consequences for the R-factor interpretation. Two distinct R factors can be computed from jet MC:

- R_{raw} (signal-contaminated): uses all jet MC counts without truth subtraction. Always > 1 (range 1.47–6.16), monotonically increasing.

- R_{bkg} ($=$ `h_R`, pure background): after subtracting truth-matched signal from each region. Crosses 1.0 at $p_T \approx 14$ GeV (range 0.675–1.854).

Table 10 compares the two.

Table 10: R-factor decomposition: signal-contaminated (R_{raw}, from jet MC totals) vs. pure background ($R_{\text{bkg}} =$ `h_R`, after truth subtraction). The earlier analysis incorrectly used R_{raw} as "pure background R."

p_T bin (GeV)	R_{raw} (contaminated)	R_{bkg} (`h_R`, pure bkg)
$[8, 10]$	1.47	0.675
$[10, 12]$	1.73	0.729
$[12, 14]$	2.29	0.904
$[14, 16]$	2.62	1.058
$[16, 18]$	3.00	1.261
$[18, 20]$	3.19	0.991
$[20, 22]$	4.21	1.273
$[22, 24]$	5.03	1.779
$[24, 26]$	3.22	1.151
$[26, 28]$	4.82	1.617
$[28, 32]$	6.16	1.854
$[32, 36]$	3.08	0.827

The factor-of-2-to-6 difference between R_{raw} and R_{bkg} is entirely due to signal contamination inflating region A. The earlier isoET overlay plots (Figure 3) used the total jet MC in the tight region, where the sharp isolation peak near 1 GeV was dominated by signal photons, not background. After truth subtraction, the background-only isoET distributions (Figure 5) show that tight and non-tight background have nearly identical isolation at low p_T (8–14 GeV), with only a subtle divergence at high p_T.

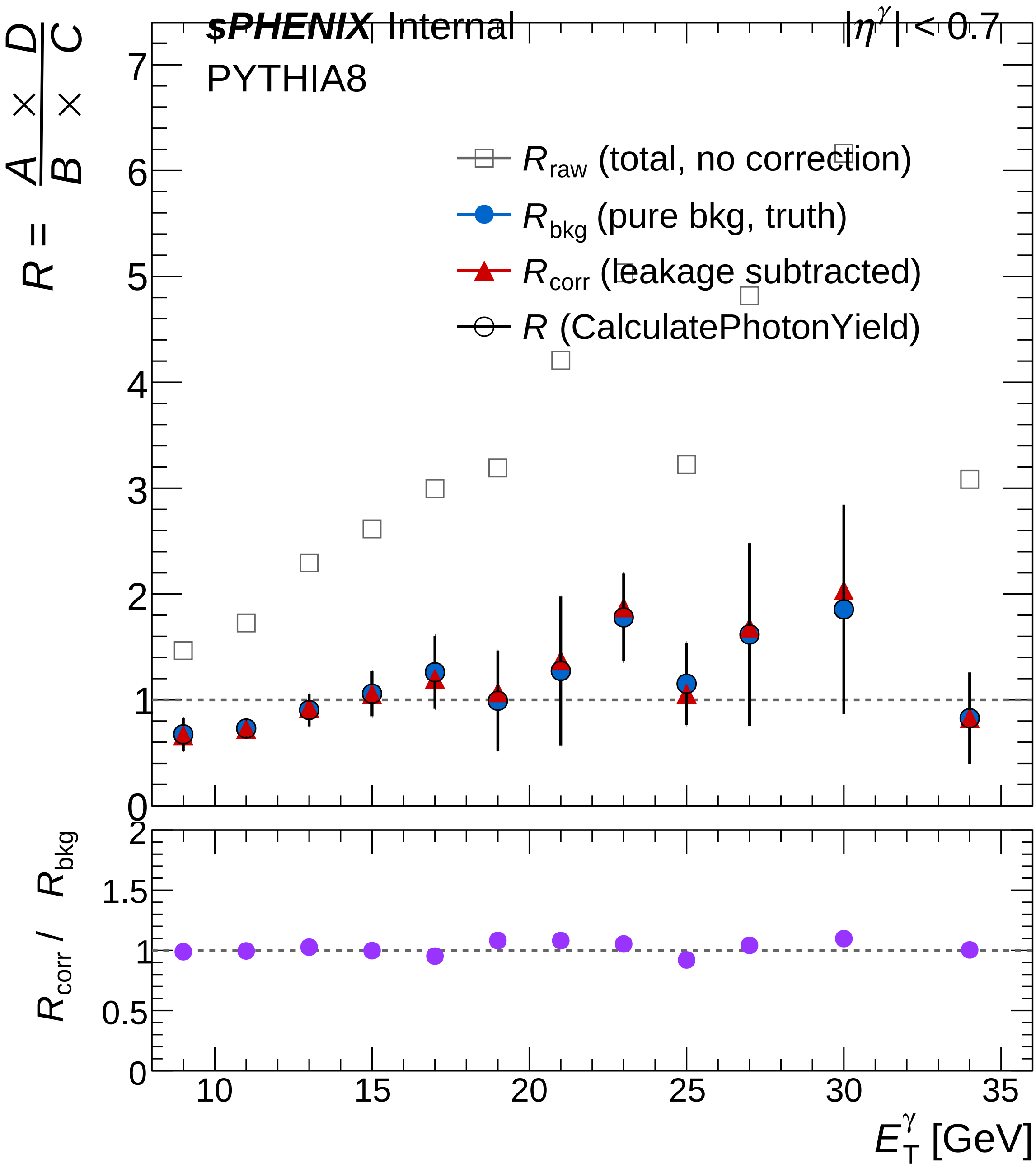

Figure 4: R-factor decomposition vs. p_T: signal-contaminated $R_{\rm raw}$ (from jet MC totals), pure background $R_{\rm bkg}$ (= `h_R`), and jet MC signal fraction in region A. Produced with `plot_R_decomposition.C`.

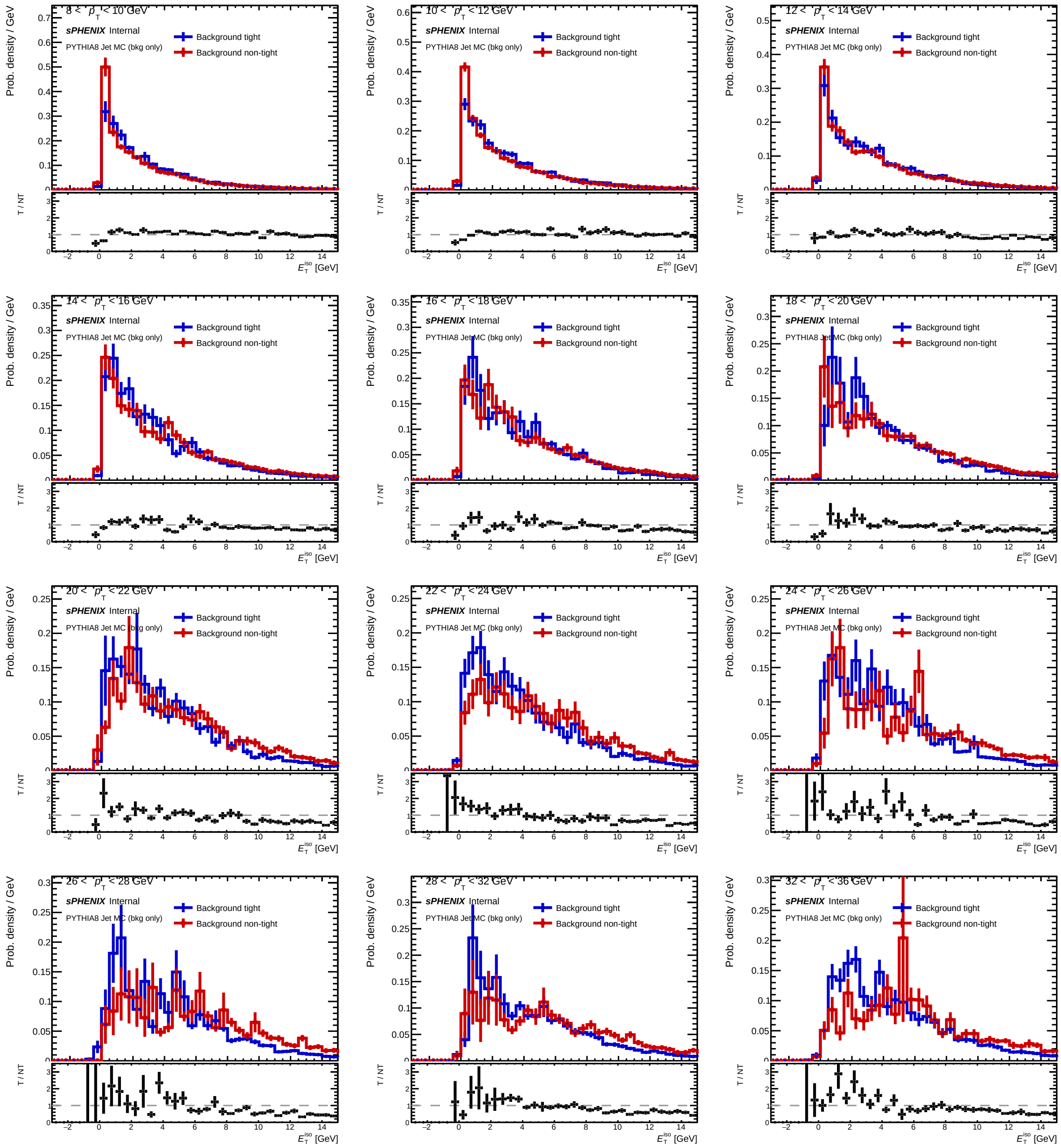

Figure 5: Normalized isolation E_T distribution for **pure background** (after truth signal subtraction) in tight and non-tight BDT regions, per p_T bin. Compare with Figure 3, which used signal-contaminated jet MC totals. Produced with `plot_bkg_isoET_tight_vs_nontight.C`.

Corrected interpretation: The R_{bkg} crossover at $p_T \approx 14$ GeV is a genuine but subtle background physics effect. At low p_T, tight and non-tight background jets have nearly identical isolation properties ($R \approx 1$, slightly below). At high p_T, photon-like background jets become marginally more isolated, pushing R above 1. The earlier conclusion that the "pure background" R is always > 1 (1.4–5.0 range) and that the sign flip in `h_R` was caused by signal leakage corrections was incorrect—those values were R_{raw} computed from signal-contaminated jet MC totals.

7.8 Updated Recommendations

In addition to the original recommendations (Section 6):

6. The purity correction sign flip is a **genuine background physics effect** (R_{bkg} crossing 1.0) that cannot be eliminated by adjusting the non-tight BDT boundary. The magnitude is modest after removing signal contamination from the diagnostics.

7. **Jet MC diagnostics must account for signal contamination**: any isolation-based ABCD diagnostic computed from raw jet MC totals is dominated by the 58–78% signal fraction in region A. Always use truth-subtracted quantities (`h_R`, background-only isoET) for physics interpretation of background behavior.

8. Approximately 30 systematic variation configs still use the old fixed non-tight bound (`bdt_max` = 0.50, missing `bdt_max_slope`/`bdt_max_intercept` keys). Regenerate if the E_T-dependent bound is intended for all variations.

9. `RecoEffCalculator.C` (old code path) does not read the parametric non-tight fields—only `RecoEffCalculator_TTreeReader.C` does. Confirm which code path is active in production.

A Plots

The following diagnostic plots were produced by the investigation agents.

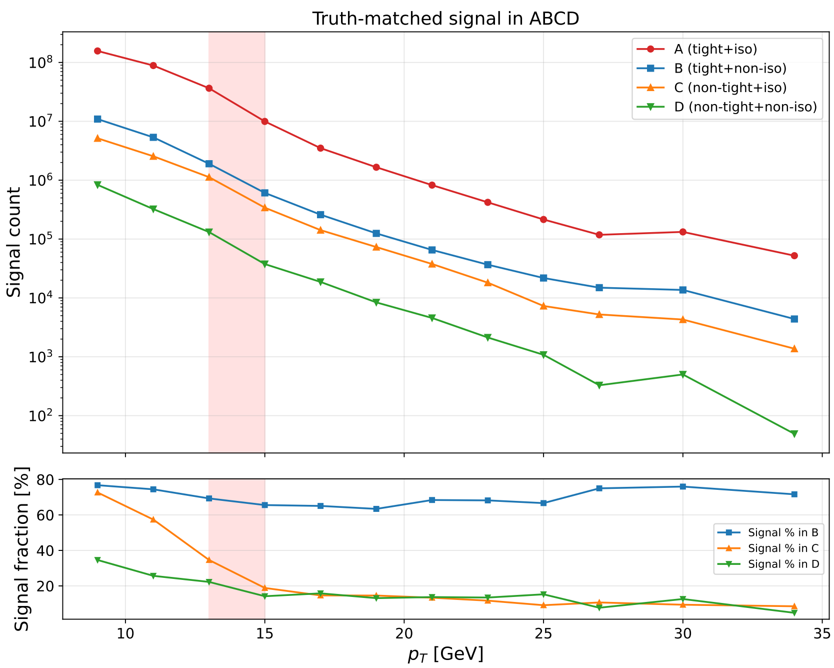

Figure 6: Signal counts in each ABCD region vs. p_T.

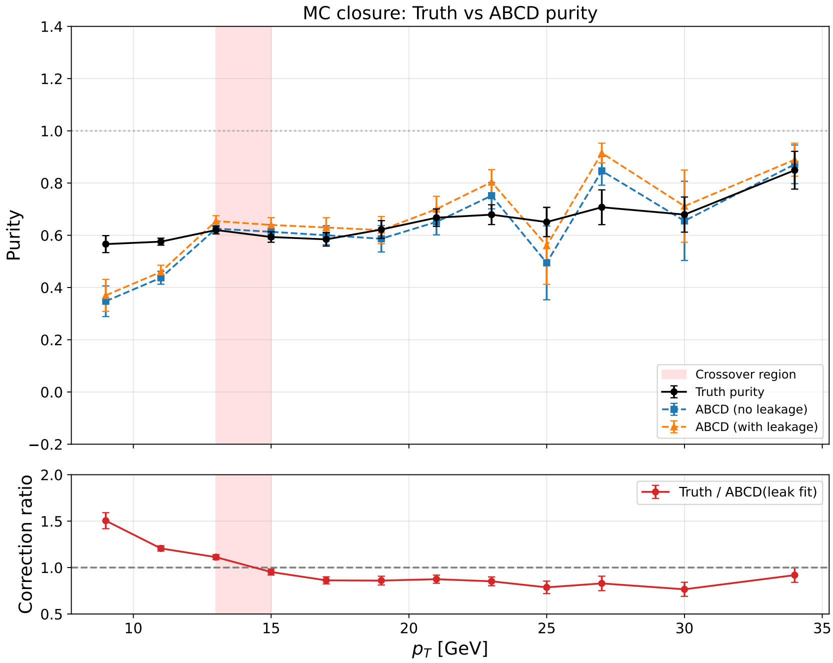

Figure 7: Truth purity vs. leakage-corrected ABCD purity as a function of p_T.

Additional multi-page diagnostics are in `plotting/figures/purity_fit_diagnostics.pdf` (Padé fit curves, residuals, ratio crossing analysis).

Note (April 8, 2026): The diagnostic plots in this appendix (Figures 6–7) were generated with the *old* fixed non-tight BDT configuration. The underlying ROOT result files have since been overwritten with the current E_T-dependent configuration. To regenerate the old plots, one would need to rerun the pipeline with `bdt_max_slope: 0` and `bdt_max_intercept: 0.5` in the non-tight block.